

Restaurant Overview Dashboard – Star Restaurant Identification & Recommendation System

1. Introduction

The rapid growth of online food delivery platforms has created intense competition among restaurants. To enhance customer experience and improve restaurant recommendations, it is essential to analyze restaurant performance using data-driven approaches.

This project focuses on analyzing restaurant data provided by a restaurant consolidator to identify star restaurants and generate intelligent recommendations. Using exploratory data analysis (EDA) and interactive dashboarding techniques, the project aims to understand restaurant behavior based on ratings, votes, cuisines, cost, online delivery, and table booking features.

The final outcome is an interactive Tableau dashboard that enables business users to explore restaurant insights effectively.

2. Problem Statement

A restaurant consolidator is planning to revamp its B2C portal using intelligent automation technology. To support this initiative, it is necessary to:

- Identify star restaurants based on performance metrics
- Analyze customer behavior and engagement
- Understand the impact of cost, cuisines, delivery, and booking features on ratings
- Generate meaningful restaurant recommendations

Two datasets were provided:

1. **Restaurant Data** – containing 19 attributes
 2. **Country Code Data** – containing country code mappings
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3. Dataset Description

Restaurant Dataset Attributes

- Restaurant ID
- Restaurant Name
- Country Code
- City
- Address
- Locality

- Cuisines
- Average Cost for Two
- Price Range
- Aggregate Rating
- Rating Text
- Votes
- Has Online Delivery
- Has Table Booking
- Latitude
- Longitude
- Currency

Country Code Dataset

- Country Code
- Country Name

Both datasets were joined using the **Country Code** field.

4. Data Understanding and Inspection

Initial Data Inspection

- Total records: 9,551 restaurants
- Total attributes: 19
- Data types included numeric, categorical, and geographical fields

Data Quality Checks

- Missing values identified in:
 - Cuisines
 - Rating Text
- Duplicate restaurant IDs were checked and removed
- Country Code dataset was validated for consistency

Data Cleaning Steps

- Removed duplicate records
- Handled null values
- Converted Yes/No fields into flags

- Verified latitude and longitude values for mapping
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5. Exploratory Data Analysis (EDA)

5.1 Geographical Distribution

- Restaurants are distributed across multiple countries
- Highest concentration observed in:
 - India
 - United States
 - United Kingdom
 - Australia

Cities such as **New Delhi, Gurgaon, and Noida** show the highest restaurant density.

5.2 Franchise Presence

Large restaurant chains exhibit strong national presence, indicating successful franchising models and brand influence across cities.

5.3 Table Booking Analysis

- Restaurants allowing table booking: **12.12%**
- Restaurants without table booking: **87.88%**

This indicates table booking is still an underutilized feature.

5.4 Online Delivery Analysis

- Restaurants providing online delivery: **25.66%**
- Restaurants without delivery: **74.34%**

Online delivery significantly contributes to customer engagement and higher votes.

5.5 Votes Comparison

Restaurants offering online delivery receive significantly higher average votes compared to non-delivery restaurants, showing higher customer interaction.

6. Cuisine Analysis

Top 10 Cuisines Served

- North Indian
- Chinese
- Fast Food
- Cafe
- Bakery
- Street Food
- Desserts
- South Indian
- Continental
- Mughlai

North Indian and Chinese cuisines dominate across cities, reflecting strong customer preference.

Cuisine Variety

- Minimum cuisines served: 1
- Maximum cuisines served: Multiple cuisines per restaurant
- Restaurants offering multiple cuisines generally attract higher customer interest.

7. Cost and Rating Analysis

Cost Distribution

Restaurants were categorized into four price ranges:

Price Range Description

1	Low cost
2	Medium
3	Premium
4	Luxury

Cost vs Rating

- Higher price range restaurants tend to receive better ratings
 - Premium and luxury restaurants show improved dining experience and service quality
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8. Ratings Analysis

Rating Distribution

- Majority of restaurants fall between ratings **2.5 – 4.0**
- Very few restaurants exceed 4.5 rating

Factors Affecting Ratings

- Number of customer votes
- Online delivery availability
- Table booking feature
- Price range
- Cuisine variety

Restaurants with high votes and service availability consistently achieve better ratings.

9. Star Restaurant Identification Logic

Star restaurants were identified using a multi-factor evaluation approach rather than ratings alone.

Factors Considered

- Aggregate rating
- Number of customer votes
- Online delivery availability
- Table booking availability
- Price range
- Cuisine diversity

Restaurants scoring highly across these parameters were classified as **Star Restaurants**, ensuring both quality and popularity are considered.

10. Dashboard Development

An interactive dashboard was developed using **Tableau Public** with the following components:

Dashboard Sections

- KPI Indicators
 - Total Restaurants
 - Average Rating
 - Online Delivery %

- Table Booking %
- Restaurant Distribution Map
- Rating vs Votes Scatter Plot
- Cost vs Rating Bar Chart
- Top 10 Cuisines Analysis

Interactive Features

- Country filter
- City filter
- Online delivery filter
- Table booking filter
- Price range filter

These filters allow users to dynamically explore restaurant insights.

11. Key Insights

- India has the highest number of restaurants in the dataset
 - Restaurants with online delivery receive higher customer engagement
 - Higher price range restaurants generally achieve better ratings
 - North Indian and Chinese cuisines dominate customer preference
 - Votes strongly correlate with restaurant visibility and ratings
 - Star restaurants combine quality, popularity, and service features
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12. Conclusion

This project successfully demonstrates how exploratory data analysis combined with interactive visualization can help identify star restaurants and generate intelligent recommendations.

The Tableau dashboard provides a comprehensive view of restaurant performance, enabling stakeholders to:

- Improve customer experience
- Enhance restaurant recommendations
- Support strategic business decisions
- Identify high-performing restaurants efficiently

The solution aligns with the objective of revamping the B2C portal using intelligent automation and data-driven insights.

13. Tools and Technologies Used

- **Python** – Data preprocessing and EDA
 - **Tableau Public** – Dashboard development
 - **Microsoft Excel** – Data inspection
 - **Mapbox / OpenStreetMap** – Geographical visualization
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14. Future Scope

- Integration of machine learning recommendation models
- Real-time data ingestion
- Customer sentiment analysis using reviews
- Personalized restaurant recommendations
- Predictive rating analysis